

Prepared for DBS Architectes · August 2026

1. Scope

Appraisal of the existing reinforced concrete frame of the former warehouse at Ateliers du Rhône, to establish capacity for the proposed change of use to studios and workshops.

2. Existing construction

The frame is in-situ reinforced concrete on a 6.0 by 7.2 m grid, built in 1962. Slabs are 180 mm one-way spanning. Columns are 400 by 400 mm. Foundations are pad bases, assumed 1.8 m square from trial holes.

3. Condition

Cover meter survey found cover between 18 and 34 mm against a nominal 25 mm. Carbonation depth averaged 11 mm. Chloride content is below the threshold. Two areas of spalling were noted at the east elevation, both associated with rainwater goods rather than systemic corrosion.

4. Capacity

Existing slab capacity is 3.0 kN/m² imposed. The proposed workshop use requires 5.0 kN/m². Strengthening is required to bays B3 to B6 and C3 to C6. Carbon fibre plate bonded to the soffit is recommended.

5. New openings

The proposed stair opening at grid B4 requires trimming. A steel frame picking up on the existing columns is recommended over slab widening.

6. Conclusions

The frame is sound and suitable for retention. Strengthening of eight bays and local repair of two spalled areas is required. Retention avoids an estimated 340 tonnes of embodied carbon against rebuild.